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ASSIGNMENT: 2

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"Properties of Determinants (All)"

What is a determinant?

- A scalar value that is calculated from the elements of a square matrix is called a determinant.
- The vertical lines are known as columns and the horizontal lines are known as rows.
- If a determinant is of order "n" then it has n number of rows and n number of columns.

Determinant of a 2x2 matrix:

Matrix:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

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Determinant of a 3x3 matrix:

Matrix:

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \Rightarrow \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

$$= a(ei-fh) - b(di-fg) - c(dh-eg)$$

Properties of Determinants:

1- There will be no change in the value of determinant if rows and columns are interchanged.

2- If rows and columns are interchanged, then its sign changes as well.

3- Same rows or columns of a determinant, (are same means same values) then the determinant will be equal to 0.

4- If any row or column of a determinant is multiplied by any variable say "k" then its value is also multiplied by k.

5- If some or all elements of a row or column are expressed as the sum of two or more terms, then the determinant can be expressed as the sum of two or more determinants.

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Detailed properties of determinants:

1- Reflection Property:

- The value of the determinant remains unchanged if rows and columns are interchanged.

Example:

$$\begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix} \Rightarrow \begin{vmatrix} 2 & 6 & 1 \\ -3 & 0 & 5 \\ 5 & 4 & -7 \end{vmatrix}$$

Determinant 1:

$= 2[(0)(-7) - (4)(5)] - (-3)[(6)(-7) - (4)(1)] + 5[(6)(5) - (1)(0)]$

$= 2[0-20] - (-3)[-42-4] + 5[30-0]$

$= 2(-20) - (-3)(-46) + 150$

$= -40 - 138 + 150$

$= -28.$

Determinant 2:

$= 2[(0)(-7) - (5)(4)] - 6[(-3)(-7) - (5)(5)] + 1[(-3)(4) - (0)(5)]$

$= 2[0-20] - 6[21-25] + 1[-12]$

$= 2(-20) - 6(-4) + 1(-12)$

$= -40 + 24 - 12$

$= -28$

Hence, the answers of both the determinants are same.

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2- **Switching Property:**

If any two rows or columns of a determinant are interchanged then sign of determinant changes as well.

Example:

$$\begin{vmatrix} 2 & -3 & 4 \\ 6 & 0 & 5 \\ 1 & 6 & -7 \end{vmatrix} \Rightarrow \text{Interchanging } R_1 \text{ and } R_3.$$

$$\begin{vmatrix} 1 & 6 & -7 \\ 6 & 0 & 5 \\ 2 & -3 & 4 \end{vmatrix} \Rightarrow \text{Resultant Determinant}$$

Determinant 1:

$= 2[(0)(-7) - (5)(6)] - (-3)[(6)(-7) - (5)(1)] + 4[(6)(6) - (0)(1)]$

$= 2[0-30] - (-3)[-42-5] + 4[36]$

$= 2(-30) - (-3)(-47) + 4(36)$

$= -60 - 141 + 144$

$= -57.$

Determinant 2:

$= 1[(0)(4) - 5(-3)] - 6[(6)(4) - (5)(2)] - 7[(6)(-3) - 2(0)]$

$= 1[0+15] - 6[24-10] - 7[-18]$

$= 1(15) - 6(14) - 7(-18)$

$= 15 - 84 + 126$

$= +57.$

Hence, the sign of determinant is changed.

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3- **All-zero Property:**

If all the elements of a column or row are zero then determinant is also 0.

Example:

$$\begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 9 & 7 & 4 \end{vmatrix} = 0$$

4- **Repetition / Proportionality Property:**

If any two rows or columns of a determinant are same, value of determinant is 0.

Example:

$$\begin{vmatrix} 4 & 6 & 4 \\ 4 & 6 & 4 \\ 9 & 6 & 4 \end{vmatrix} = 0$$

5- **Scalar Multiple Property:**

If the elements of a row or column of a determinant are multiplied by any non-zero constant then the determinant also gets multiplied by the same constant.

Example:

$$\begin{vmatrix} 102 & 18 & 36 \\ 13 & 4 & 1 \\ 17 & 3 & 6 \end{vmatrix} = \begin{vmatrix} 6(17) & 6(3) & 6(6) \\ 13 & 4 & 1 \\ 17 & 3 & 6 \end{vmatrix}$$

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$$= 6 \times \begin{vmatrix} 17 & 3 & 6 \\ 13 & 4 & 1 \\ 17 & 3 & 6 \end{vmatrix}$$

6- **Sum Property:**

If elements of a row or column of a determinant are expressed as sum of two (or more) terms, then the determinant can be expressed as sum of (or more) determinants.

Example:

$$\begin{vmatrix} a & b & c \\ a+2x & b+2y & c+2z \\ x & y & z \end{vmatrix} = \begin{vmatrix} a & b & c \\ a & b & c \\ x & y & z \end{vmatrix} + \begin{vmatrix} a & b & c \\ 2x & 2y & 2z \\ x & y & z \end{vmatrix}$$

7- **Property of invariance:**

Suppose any scalar multiples of corresponding elements of other rows or columns are added to every element of any row or column of a determinant. In this case, the value of determinant remains same.

Example:

$$\Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = \begin{vmatrix} a_1+kb_1 & a_2+kb_2 & a_3+kb_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

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8- **Factor Property:**

If a determinant Δ becomes zero when we insert $x = \alpha$ then $(x - \alpha)$ is a factor of Δ. This is called the factor property.

Example:

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \Rightarrow \Delta 1 = \begin{vmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{vmatrix}$$

where C_{ij} denotes the cofactor of the element a_{ij} in Δ.

9- **Triangle Property:**

If the elements of a determinant above or below the main diagonal consists of zero then the determinant is equal to the product of diagonal elements.

Example:

$$\begin{vmatrix} 4 & 5 & 1 \\ 0 & 9 & 2 \\ 0 & 0 & 3 \end{vmatrix} = \begin{vmatrix} 4 & 0 & 0 \\ 5 & 9 & 0 \\ 1 & 2 & 3 \end{vmatrix} = (4)(9)(3) = 108$$

10- **Determinant of co-factor matrix:**

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}, \quad \Delta 1 = \begin{vmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{vmatrix}$$

where C_{ij} denotes the co-factors of

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the elements a_{ij} in Δ.

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